

Tutorial Sheet - 4

PART - A

Q. 1 Define first order first degree ordinary differential equation.

Ans A first order first degree differential equation is an equation involving the first derivative only and the highest power of the derivative is 1.

General form :- $f(x, y, \frac{dy}{dx}) = 0$ or $\frac{dy}{dx} = f(x, y)$

Q. 2 Find the order and degree of $p^2 + 2py \cot x = y^2$

Solⁿ Here $p = \frac{dy}{dx}$

$$\Rightarrow \left(\frac{dy}{dx}\right)^2 + 2y \cot x \left(\frac{dy}{dx}\right) = y^2$$

• Highest derivative = $\frac{dy}{dx}$

• Highest power of derivative = 2

Order = 1

Degree = 2

Q. 3 Solve the differential equation $y = px + a/p$.

Solⁿ Here $p = \frac{dy}{dx}$

This is Clairaut's equation.

General form : $y = px + f(p)$

Here, $f(p) = \frac{a}{p}$

substitute $p = C \Rightarrow y = Cx + \frac{a}{C}$

Q.4 solve the differential equation $p^2 - 2p - 3 = 0$

Solⁿ

$$p^2 - 2p - 3 \Rightarrow p^2 - 3p + p - 3 = 0$$

$$p(p-3) + 1(p-3) = 0$$

$$(p+1)(p-3) = 0$$

$$p = 3 \text{ or } p = -1$$

$$p = \frac{dy}{dx}$$

Case 1:- $\frac{dy}{dx} = 3$, On Integration

$$y = 3x + C_1$$

Case 2:- $\frac{dy}{dx} = -1$, On Integration

$$y = -x + C_2$$

$$\Rightarrow (x - y - C_1)(y - 3x - C_2) = 0 \quad \text{Ans}$$

Q.5 What is Clairaut's Equation?

Ans A differential equation of the form $y = px + f(p)$

where $p = \frac{dy}{dx}$ is called Clairaut's differential equation.

General solution: $y = cx + f(c)$

PART - B

Q.1 Solve the differential equation $x^2 p^2 - 2xy p + 2y^2 - x^2 = 0$.

Solⁿ Here $p = \frac{dy}{dx}$

Divide by $x^2 \Rightarrow p^2 - \frac{2y}{x} p + 2 \left(\frac{y}{x}\right)^2 - 1 = 0$

Let $v = \frac{y}{x}$, So, $y = vx$

Differentiate $p = \frac{dy}{dx} = v + x \frac{dv}{dx}$

Substitute

After simplification we obtain $\frac{dv}{\sqrt{1-v^2}} = \frac{dx}{x}$

Integrate

$$\sin^{-1} v = \ln x + C$$

Substitute $v = \frac{y}{x}$

$$\sin^{-1} (y/x) = \log cx \quad \text{Ans}$$

Q.2 Solve the differential equation $y + px = p^2 x^4$.

Solⁿ $y = x p^2 x^2 - px$

This is Lagrange's equation :- $y = xp + f(p)$

Differentiate :- $p = p + x \frac{dp}{dx} + f'(p) \frac{dp}{dx}$

Solving the resulting differential equation and integrate

we get , $xy = c^2 x - c$ Ans

Q.3 Solve the differential equation $y = 2px + y^2 p^3$.

Solⁿ Rearrange $\rightarrow 2x = \frac{y - y^2 p^3}{p} = \frac{y}{p} - y^2 p^3$

Differentiate w.r.t y

$2 \frac{1}{p} = \frac{d}{dy} \left(\frac{y}{p} - y^2 p^3 \right)$ { $dx/dy = 1/p$ }

$\Rightarrow \frac{2}{p} = \left(\frac{1}{p} - \frac{y}{p^2} \frac{dp}{dy} \right) - (2yp^2 + 2y^2 p \frac{dp}{dy})$

$\Rightarrow \frac{2}{p} - \frac{1}{p} + 2yp^2 + \left(\frac{y}{p^2} + 2y^2 p \right) \frac{dp}{dy} = 0$

$\frac{1}{p} + 2yp^2 + \frac{y}{p^2} (1 + 2yp^3) \frac{dp}{dy} = 0$

$\frac{1 + 2yp^3}{p} + \frac{y}{p^2} (1 + 2yp^3) \frac{dp}{dy} = 0$

$$(1 + 2yp^3) \left(\frac{1}{p} + \frac{y}{p^2} \frac{dp}{dy} \right) = 0$$

Solve the linear ODE

$$\Rightarrow \frac{1}{p} + \frac{y}{p^2} \frac{dp}{dy} = 0 \rightarrow \frac{1}{y} dy + \frac{1}{p} dp = 0$$

Integrating both side

$$\ln(y) + \ln(p) = \ln(C) \Rightarrow yp = C \Rightarrow p = \frac{C}{y}$$

Substitute $p = \frac{C}{y}$ in original eqⁿ

$$y = 2 \left(\frac{C}{y} \right) x + y^2 \left(\frac{C}{y} \right)^3$$

$$y = \frac{2Cx}{y} + \frac{C^3}{y} \Rightarrow y^2 = 2Cx + C^3 \quad \text{Ans}$$

PART - C

Q.2 Solve the differential equation $p^2 + 2py \cot x = y^2$

Solⁿ Rewrite :- $(p + y \cot x)^2 = y^2 (1 + \cot^2 x)$
 $(p + y \cot x)^2 = y^2 \operatorname{cosec}^2 x$

Take square root :- $p + y \cot x = \pm y \operatorname{cosec} x$

So,

$$p = y (\pm \operatorname{cosec} x - \cot x)$$

This gives a separable equation, integration leads to

$$y = \frac{C_1}{1 + \cos x} = 0 \quad \text{or} \quad y = \frac{C_2}{1 - \cos x} = 0$$

$$\left(y - \frac{C_1}{1 + \cos x} \right) \left(y - \frac{C_2}{1 - \cos x} \right) = 0$$

Q.2 Solve the differential equation $p^3 - 4xyp + 8y^2 = 0$.

Solⁿ Rearrange :- $4xyp = p^3 + 8y^2$

Dividing by $4yp$:- $x = \frac{p^2}{4y} + \frac{2y}{p}$

Diff. w.r.t y

$$\frac{1}{p} = \frac{d}{dy} \left(\frac{p^2}{4y} + \frac{2y}{p} \right) \quad \left\{ \frac{dx}{dy} = \frac{1}{p} \right\}$$

$$\frac{1}{p} = \left[\frac{1}{4y} \left(2p \frac{dp}{dy} \right) - \frac{p^2}{4y^2} \right] + \left[\frac{2}{p} - \frac{2y}{p^2} \frac{dp}{dy} \right]$$

$$\Rightarrow \frac{1}{p} - \frac{2}{p} + \frac{p^2}{4y^2} = \left(\frac{2p}{4y} - \frac{2y}{p^2} \right) \frac{dp}{dy}$$

$$-\frac{1}{p} + \frac{p^2}{4y^2} = \left(\frac{p}{2y} - \frac{2y}{p^2} \right) \frac{dp}{dy}$$

$$\Rightarrow \frac{p^3 - 4y^2}{4py^2} = \left(\frac{p^3 - 4y^2}{2yp^2} \right) \frac{dp}{dy}$$

Solve the ODE.

$$\frac{1}{2y} = \frac{1}{p} \frac{dp}{dy} \Rightarrow \frac{dp}{p} = \frac{dy}{2y}$$

Integrating $\Rightarrow \ln(p) = \frac{1}{2} \ln y + \ln c$

$$\ln p = \ln(c\sqrt{y}) \Rightarrow p = c\sqrt{y}$$

$$\Rightarrow p^2 = c^2 y$$

Substitute $p^2 = c^2 y$ into the original eqⁿ

$$(cy^{1/2})^3 - 4xy(cy^{1/2}) + 8y^2 = 0$$

$$c^3 y^{3/2} - 4cxy^{3/2} + 8y^2 = 0$$

{ Divide by $y^{3/2}$ (assuming $y \neq 0$) }

$$c^3 - 4cx + 8\sqrt{y} = 0$$

$$8\sqrt{y} = 4cx - c^3$$

{ Square both sides }

$$64y = (4cx - c^3)^2$$

$$64y = c^2(4x - c^2)^2$$

Q. 3

Solve the differential equation $(x-a)p^2 + (x-y)p - y = 0$.

Solⁿ

$$(x-a)p^2 + xp - yp - y = 0$$

$$(x-a)p^2 + xp = yp + y$$

$$xp^2 - ap^2 + xp = y(p+1)$$

soln for y $\Rightarrow y = \frac{xp^2 + xp - ap^2}{p+1}$

$$y = \frac{xp(p+1) - ap^2}{p+1}$$

$$y = \frac{xp(p+1)}{p+1} - \frac{ap^2}{p+1}$$

$$y = px - \frac{ap^2}{p+1}$$

Equation is Clairaut's Eqⁿ

So $y = cx - \frac{ac^2}{c+1}$

~~Ans~~
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